

NET512

Designing and Implementing a Secure, Scalable, and Automated Enterprise Network

Weekly Practical Plan — NQF Level 5 | 17 Credits

Week 5: Cisco Router and Switch Basics (10 Marks)

28.

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1
R1(config)#
```

29.

```
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
```

30.

```
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1
SW1(config)#
```

31.

```
R1#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.1	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```
R1#
```

32.

R2#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.2	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

R2#

33.

SW1#show ip interface brief

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/1	unassigned	YES	manual	up	up
FastEthernet0/2	unassigned	YES	manual	up	up
FastEthernet0/3	unassigned	YES	manual	down	down
FastEthernet0/4	unassigned	YES	manual	down	down
FastEthernet0/5	unassigned	YES	manual	down	down
FastEthernet0/6	unassigned	YES	manual	down	down
FastEthernet0/7	unassigned	YES	manual	down	down
FastEthernet0/8	unassigned	YES	manual	down	down
FastEthernet0/9	unassigned	YES	manual	down	down
FastEthernet0/10	unassigned	YES	manual	down	down
FastEthernet0/11	unassigned	YES	manual	down	down
FastEthernet0/12	unassigned	YES	manual	down	down
FastEthernet0/13	unassigned	YES	manual	down	down
FastEthernet0/14	unassigned	YES	manual	down	down
FastEthernet0/15	unassigned	YES	manual	down	down
FastEthernet0/16	unassigned	YES	manual	down	down
FastEthernet0/17	unassigned	YES	manual	down	down
FastEthernet0/18	unassigned	YES	manual	down	down
FastEthernet0/19	unassigned	YES	manual	down	down
FastEthernet0/20	unassigned	YES	manual	down	down
FastEthernet0/21	unassigned	YES	manual	down	down
FastEthernet0/22	unassigned	YES	manual	down	down
FastEthernet0/23	unassigned	YES	manual	down	down
FastEthernet0/24	unassigned	YES	manual	down	down
GigabitEthernet0/1	unassigned	YES	manual	down	down
GigabitEthernet0/2	unassigned	YES	manual	down	down
Vlan1	10.10.10.10	YES	manual	up	up

SW1#

34.

SW1#ping 10.10.10.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.10.10.2, timeout is 2 seconds:

.....

Success rate is 80 percent (4/5), round-trip min/avg/max = 0/2/11 ms

SW1#

```
!
interface Vlan1
  ip address 10.10.10.10 255.255.255.0
!
ip default-gateway 10.10.10.2
!
!
!
line con 0
!
line vty 0 4
  login
line vty 5 15
  login
!
!
!
!
end
```

35.

```

R1#show interfaces
FastEthernet0/0 is up, line protocol is up (connected)
  Hardware is Lance, address is 0060.2fa9.1401 (bia 0060.2fa9.1401)
  Description: Link to SW1
  Internet address is 10.10.10.1/24
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
FastEthernet0/1 is administratively down, line protocol is down (disabled)
  Hardware is Lance, address is 0060.2fa9.1402 (bia 0060.2fa9.1402)
  MTU 1500 bytes, BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Full-duplex, 100Mb/s, media type is RJ45
  ARP type: ARPA, ARP Timeout 04:00:00,
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0 (size/max/drops); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    0 packets input, 0 bytes, 0 no buffer
    Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 input packets with dribble condition detected
    0 packets output, 0 bytes, 0 underruns
    0 output errors, 0 collisions, 1 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out
Vlan1 is administratively down, line protocol is down
  Hardware is CPU Interface, address is 000c.cfb7.26e7 (bia 000c.cfb7.26e7)
  MTU 1500 bytes, BW 100000 Kbit, DLY 1000000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 21:40:21, output never, output hang never

```

```

R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability    Platform    Port ID
SW1          Fas 0/0          177        S             2960        Fas 0/1
R1#show cdp neighbors detail

Device ID: SW1
Entry address(es):
  IP address : 10.10.10.10
Platform: cisco 2960, Capabilities: Switch
Interface: FastEthernet0/0, Port ID (outgoing port): FastEthernet0/1
Holdtime: 160

Version :
Cisco IOS Software, C2960 Software (C2960-LANBASEK9-M), Version 15.0(2)SE4, RELEASE SOFTWARE (fc1)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2013 by Cisco Systems, Inc.
Compiled Wed 26-Jun-13 02:49 by mnnguyen

advertisement version: 2
Duplex: full
...
```

```

SW1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID    Local Intrfce    Holdtme    Capability    Platform    Port ID
R1           Fas 0/1          133        R             C2800        Fas 0/0
R2           Fas 0/2          167        R             C2800        Fas 0/0
SW1#
```

37.

```

R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface fastethernet 0/0
R1(config-if)#no cdp enable
R1(config-if)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show cdp interface
Vlan1 is administratively down, line protocol is down
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
FastEthernet0/1 is administratively down, line protocol is down
  Sending CDP packets every 60 seconds
  Holdtime is 180 seconds
R1#
```

38.

```

R1#config t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#no cdp run
R1(config)#cdp run
R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
```

39.

```

R1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone
Device ID      Local Intrfce  Holdtme   Capability   Platform   Port ID
R1#

```

40.

```

SW1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES manual up          up
FastEthernet0/2 unassigned      YES manual up          up
FastEthernet0/3 unassigned      YES manual down        down
FastEthernet0/4 unassigned      YES manual down        down
FastEthernet0/5 unassigned      YES manual down        down
FastEthernet0/6 unassigned      YES manual down        down
FastEthernet0/7 unassigned      YES manual down        down
FastEthernet0/8 unassigned      YES manual down        down
FastEthernet0/9 unassigned      YES manual down        down
FastEthernet0/10 unassigned      YES manual down        down
FastEthernet0/11 unassigned      YES manual down        down
FastEthernet0/12 unassigned      YES manual down        down
FastEthernet0/13 unassigned      YES manual down        down
FastEthernet0/14 unassigned      YES manual down        down
FastEthernet0/15 unassigned      YES manual down        down
FastEthernet0/16 unassigned      YES manual down        down
FastEthernet0/17 unassigned      YES manual down        down
FastEthernet0/18 unassigned      YES manual down        down
FastEthernet0/19 unassigned      YES manual down        down
FastEthernet0/20 unassigned      YES manual down        down
FastEthernet0/21 unassigned      YES manual down        down
--More--

```

41.

```

SW1#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/1 unassigned      YES manual up          up
FastEthernet0/2 unassigned      YES manual administratively down down
FastEthernet0/3 unassigned      YES manual down        down
FastEthernet0/4 unassigned      YES manual down        down
FastEthernet0/5 unassigned      YES manual down        down
FastEthernet0/6 unassigned      YES manual down        down
FastEthernet0/7 unassigned      YES manual down        down
FastEthernet0/8 unassigned      YES manual down        down
FastEthernet0/9 unassigned      YES manual down        down
FastEthernet0/10 unassigned      YES manual down        down
FastEthernet0/11 unassigned      YES manual down        down
FastEthernet0/12 unassigned      YES manual down        down
FastEthernet0/13 unassigned      YES manual down        down
FastEthernet0/14 unassigned      YES manual down        down
FastEthernet0/15 unassigned      YES manual down        down
FastEthernet0/16 unassigned      YES manual down        down
FastEthernet0/17 unassigned      YES manual down        down
FastEthernet0/18 unassigned      YES manual down        down
FastEthernet0/19 unassigned      YES manual down        down
FastEthernet0/20 unassigned      YES manual down        down
FastEthernet0/21 unassigned      YES manual down        down
--More--

```

42.

```

SW1#show interfaces status
Port      Name      Status      Vlan      Duplex  Speed Type
Fa0/1     connected  1           a-full    a-100   10/100BaseTX
Fa0/2     connected  1           a-full    a-100   10/100BaseTX
Fa0/3     notconnect 1           auto      auto    10/100BaseTX
Fa0/4     notconnect 1           auto      auto    10/100BaseTX
Fa0/5     notconnect 1           auto      auto    10/100BaseTX
Fa0/6     notconnect 1           auto      auto    10/100BaseTX
Fa0/7     notconnect 1           auto      auto    10/100BaseTX
Fa0/8     notconnect 1           auto      auto    10/100BaseTX
Fa0/9     notconnect 1           auto      auto    10/100BaseTX
Fa0/10    notconnect 1           auto      auto    10/100BaseTX
Fa0/11    notconnect 1           auto      auto    10/100BaseTX
Fa0/12    notconnect 1           auto      auto    10/100BaseTX
Fa0/13    notconnect 1           auto      auto    10/100BaseTX
Fa0/14    notconnect 1           auto      auto    10/100BaseTX
Fa0/15    notconnect 1           auto      auto    10/100BaseTX
Fa0/16    notconnect 1           auto      auto    10/100BaseTX
Fa0/17    notconnect 1           auto      auto    10/100BaseTX
Fa0/18    notconnect 1           auto      auto    10/100BaseTX
Fa0/19    notconnect 1           auto      auto    10/100BaseTX
Fa0/20    notconnect 1           auto      auto    10/100BaseTX
Fa0/21    notconnect 1           auto      auto    10/100BaseTX
--More--

```

43.

```

SW1#show interfaces fastethernet 0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.c790.4102 (bia 0001.c790.4102)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Half-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out

```

44.

```

SW1#show interface fastethernet 0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.c790.4102 (bia 0001.c790.4102)
  BW 100000 Kbit, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out

```

45.

```

SW1#show interface fastethernet 0/2
FastEthernet0/2 is up, line protocol is up (connected)
  Hardware is Lance, address is 0001.c790.4102 (bia 0001.c790.4102)
  BW 100000 Kbit, DLY 1000 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 10Mb/s
  input flow-control is off, output flow-control is off
  ARP type: ARPA, ARP Timeout 00:04:00
  Last input 00:00:08, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue :0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    956 packets input, 193351 bytes, 0 no buffer
    Received 956 broadcasts, 0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
    0 watchdog, 0 multicast, 0 pause input
    0 input packets with dribble condition detected
    2357 packets output, 263570 bytes, 0 underruns
    0 output errors, 0 collisions, 10 interface resets
    0 babbles, 0 late collision, 0 deferred
    0 lost carrier, 0 no carrier
    0 output buffer failures, 0 output buffers swapped out

```

46.

```

R2#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.10.10.2	YES	manual	up	up
FastEthernet0/1	unassigned	YES	unset	administratively down	down
Vlan1	unassigned	YES	unset	administratively down	down

```

R2#

```

47.

When the duplex settings on SW1 and R2 are not the same, the link may be up, but the communications performance will be poor. Symptoms of duplex mismatch include loss of packets, poor network performance, retransmissions, and interface errors such as collisions or late collisions.

A duplex mismatch happens when one device is operating in half-duplex while the other is running in full-duplex mode. The half-duplex device sends signals using the CSMA/CD protocol, which assumes shared media such as hubs. It also has collision detection and recovery functions. On the other hand, a full-duplex device can send and receive data simultaneously without collision detection or recovery mechanisms. Therefore, the two devices cannot communicate efficiently, which causes errors, re-transmissions and poor network performance.